

deep-forensics

BOB-124 — Deep Forensic Investigation: 5th Forced-Logout (2026-08-19 15:28:22 CEST)

Revision: 1 **Last modified:** 2026-08-19T15:45:00Z **Verdict (§11.4.6):** SIGKILL INITIATOR UNATTRIBUTED — pattern MATCHES #2/#3/#4 signature for every observable class; kernel-audit attribution still impossible (auditd never installed). **Iron Law:** Investigation only — no fixes, no source changes.

Executive summary

user@1000.service was SIGKILLed at 2026-08-19 15:28:22 CEST exactly 20 min 41 s after login (15:07:41 → 15:28:22). The kill signature is **identical in shape** to prior incidents #2/#3/#4: a pam_tcb(gdm-password:session): Session closed for milosvasic on tty2 ses=4 fires from gdm-session-worker[3976] at the SAME journal second as the Main process exited, code=killed, status=9/KILL on user@1000.service, followed by a mass SIGKILL cascade of every child under the cgroup. No preceding systemctl stop, loginctl terminate-user, Manager.StopUnit, or KillUnit D-Bus call appears in the journal window (docs/qa/BOB-124/evidence/stop_terminate_paths.log lists only Removed session 1/4 from systemd-logind *after* the SIGKILL). Every kernel-side failure class remains ruled OUT: **no** oom-kill in the kernel journal (evidence/kernel_oom_scan.log empty), **no** systemd-oomd trigger (evidence/oomd_journal.log shows only the boot-time Start), **no** suspend/hibernate, **no** power event. Linger=yes is set (evidence/loginctl_user_state.txt) — which should preserve user@1000.service across session teardown, yet did not. This is the fifth consecutive incident where Phase 1 root-cause investigation cannot complete

because auditd/auditctl/ausearch are not installed and /etc/audit/ does not exist (evidence/loginctl_user_state.txt bottom lines). Correlate signal: user@1000.service: Consumed 1h 21.484s CPU time, 23.6G memory peak, 488.2M memory swap peak in 20 min 41 s — a memory-accumulation rate roughly ×10 faster than the prior session (23.6 GB / 21 min vs 25.1 GB / 3 h 22 min reported in docs/incidents/2026-08-19-5th-forced-logout.md), suggesting the reboot did not reset the runaway consumer. The mechanism direction hardens: something INSIDE user@1000 pushes memory into the multi-GB range while sending a gdm-password session-close on tty2, and the SIGKILL falls on the user manager PID milliseconds later — but no in-scope observer can name the initiator.

Timeline (evidence-cited)

Time (CEST)	Event	Evidence
15:06:20–35	Prior session clean shutdown (SERVICE_STOP res=success) — operator’s intentional reboot	docs/incidents/2026-08-19-5th-forced-logout.md §“Distinction from the 15:06 clean shutdown”
15:07:39–40	Host boot; swap activated (16 GB nvme + 32 GB zram); systemd-oomd.service started	docs/qa/B0B-124/evidence/oom_extended.log lines 3–15
15:07:41	user@1000.service started for milosvasic	evidence/user1000_grepped.log lines 1–2
15:07:40	milosvasic session on tty2 (ses=4) — see loginctl_user_state.txt Timestamp=Wed 2026-08-19 15:07:40 CEST	evidence/loginctl_user_state.txt
15:07:41 → 15:28:21	20 min 41 s of active work; wave-6 subagents Q/R/S/T dispatched at 15:12 (per .superpowers/sdd/progress.md); scaling_probe pytest hits saturate the merge-search	.superpowers/sdd/progress.md “Wave-6 in flight — 2026-08-19 15:12”; journal_inc5.log 15:28:00-01 batches of ratelimit 10/min exceeded

Time (CEST)	Event	Evidence
	rate limiter (many 429 Too Many Requests in journal_inc5.log)	
15:27:14 → 15:28:15	overlayfs xino warnings + crun-buildah scopes: container build activity (BOB-096 tests reload)	evidence/ kernel_events.log
15:28:08 → 15:28:19	helix-postgres errors: relation "background_tasks" does not exist (unrelated helix-server churn), then unexpected EOF on client connection with an open transaction	evidence/ pre_kill_15s_window.log
15:28:22.181	audit type=1106: op=PAM:session_close ... acct="milosvasic" exe="/ usr/libexec/gdm-session- worker" ... terminal=/ dev/tty2 res=success from gdm-session- worker[3976]	evidence/kill_moment.log lines 6996–7002
15:28:22	systemd[1]: user@1000.service: Main process exited, code=killed, status=9/ KILL	evidence/kill_moment.log line 7005
15:28:22	Mass SIGKILL cascade — 90+ children including postgres, qbittorrent-nox, webui-bridge, tmux: server, claude, python3, MainThread, hook-linux- amd64, lumen-linux-amd, chrome-devtools, caddy, jetbrainsd, yandex_browser × 15	evidence/ systemd_pid1_prekill.log
15:28:23	user@1000.service: Failed with result 'signal' /	

Time (CEST)	Event	Evidence
	Consumed 1h 21.484s CPU time, 23.6G memory peak, 488.2M memory swap peak	evidence/ user1000_grepped.log bottom
15:28:22	systemd-logind[1296]: Removed session 1 / Removed session 4 — AFTER the SIGKILL, not before	evidence/ stop_terminate_paths.log
15:28:22	GDM greeter reappears on tty1 (gdm-launch-environment ... session opened for gdm-greeter)	evidence/ gdm_compositor_events.log
15:28:35	milosvasic re-authenticated; session-7 (wayland/class=user) + session-8 (manager); user@1000 started again	evidence/ stop_terminate_paths.log line 7478; evidence/ user1000_grepped.log last two lines

Elapsed dead: ≈ 13 s (15:28:22 $\rightarrow$ 15:28:35 relogin). GDM auto-reset the greeter within 1 s.

Comparative signature table across #2 / #3 / #4 / #5

Sources: docs/incidents/2026-08-18-perceived-forced-logout-2nd.md (#2), docs/incidents/2026-08-18-3rd-forced-logout.md (#3), docs/incidents/2026-08-19-4th-forced-logout.md + docs/qa/B0B-123/incident-4-forensics.log (#4), THIS file (#5).

Signature	#2 (2026-08-18 20:50)	#3 (2026-08-18 23:45)	#4 (2026-08-19 00:37)	#5 (2026-08-19 00:37)
gdm-session-worker PAM session_close acct=milosvasic at exact SIGKILL timestamp	YES (per #4 doc)	YES (per #4 doc)	YES (inc4_forensics_head.txt line 3116)	YES (evidence/ kill_moment.log line 3116)
	YES	YES	YES	YES

Signature	#2 (2026-08-18 20:50)	#3 (2026-08-18 23:45)	#4 (2026-08-19 00:37)	#5 (2026-08-19 00:37)
terminal=/dev/tty2 res=success in the audit1106 user@1000.service: Main process exited, code=killed, status=9/KILL (bare, no preceding Stopping/ Deactivating)	YES	YES	YES	YES (evidence systemd_p
Explicit loginctl terminate / systemctl stop user@1000 / Manager.StopUnit / KillUnit in window	NO (per #4 §“Journal search ... returns EMPTY”)	NO	NO	NO (evidence stop_term only <i>post</i> 1/4)
Linger=yes (should have preserved user@1000)	YES (per #4)	YES	YES (inc4_forensics_head.txt §3)	YES (evidence loginctl_
IdleAction=ignore (verified in #4)	verified	verified	verified	not re-ch — /etc/s not read (i reproduc
Kernel OOM (oom- kill, invoked oom- killer) fired	NO	NO	NO	NO (evidence kernel_oc
systemd-oomd triggered a kill decision	NO	NO	NO	NO (evidence oomd_jour Start at 1
Suspend/hibernate/ power-state event	NO	NO	NO	NO (evidence power_sta lines abou keybindin transition
KillUserProcesses runtime state	unset (per #4)	unset	unset (per #4 §“Ruled out”) not recorded in #4 doc	not re-qu run this i

Signature	#2 (2026-08-18 20:50)	#3 (2026-08-18 23:45)	#4 (2026-08-19 00:37)	#5 (2026-08-19 00:37)
user@1000 memory peak at teardown	not recorded per-incident	not recorded		23.6 GB i (evidence user1000_ — actually user1000_)
user@1000 CPU peak at teardown	—	—	—	1 h 21.48 (≈ 4 core)
Load average at incident window	not captured	not captured	0.20, 0.70, 1.06 (light)	0.80, 5.0 (docs/qa/ forensics)
Parallel subagent load at kill	3+ concurrent (per #4 §“defensive move”)	3+	3+ (Task 85/86/BOB-122/BOB-118 landed)	4 concurr at 15:12) pytest fle
auditctl / ausearch / etc / audit/rules.d / present	NO	NO	NO (per #4 §“Recommended next step”)	NO (audi etc/audit
Session dead-time before relogin	—	—	13 min	13 s (faste

Signature match ratio: every checkable dimension for #5 matches #4 exactly. The two “not re-checked” cells (IdleAction, KillUserProcesses) were verified stable in #4 and no reboot changed logind.conf — but evidence/loginctl_user_state.txt shows IdleHint=no IdleSinceHint=0, i.e. logind considered the session non-idle at teardown.

Session context (§11.4.6, what the session was doing)

Between 15:12 and 15:28 the SDD orchestration dispatched **Wave-6** — four concurrent Claude subagents (Agents Q/R/S/T per .superpowers/sdd/progress.md) plus the merge-search scaling_probe pytest fleet was hammering /api/v1/search (journal_inc5.log at 15:28:00-01 shows ≥12 different scaling_probe_* query ids being fanned across trackers, tripping the 10/min rate limiter — 429 Too Many Requests bursts).

Agent S completed BOB-096 chaos tests at commit 76d6b1f6; other Wave-6 agents were mid-flight. `crun-buildah` scopes at 15:28:10-15 indicate container build/reload activity. `pgrep -af pytest` at investigation time returned empty (post-kill), consistent with the SIGKILL cascade having reaped them. This matches the #4 forensic note that “all 4 incidents occurred during heavy parallel subagent load” — #5 makes it 5-of-5.

The load-vs-memory correlation: `user@1000` consumed 23.6 GB peak in 20:41 elapsed. The prior session (per `docs/incidents/2026-08-19-5th-forced-logout.md`) consumed 25.1 GB in 3 h 22 min. The per-minute rate for the killed session is $\approx 1.14 \text{ GB/min}$ vs $\approx 0.124 \text{ GB/min}$ for the prior — an order of magnitude faster. Whether this rate is causal (systemd-oomd on some ancestor slice, some resource-cap watchdog, or a memory-triggered PAM-close policy) remains **§11.4.6 UNCONFIRMED**: no oomd/OOM/pressure event fired, `/proc/pressure/memory` at investigation time shows `avg60=0.00`, and the initiator is not self-attributed anywhere in the journal.

What ruled OUT this incident (evidence/cited)

- **Kernel OOM-killer:** `evidence/kernel_oom_scan.log` returns 0 hits across 15:07-15:29
 - **systemd-oomd:** `evidence/oomd_journal.log` shows only boot-time Start; no kill decision, no memory-pressure log
 - **Suspend / hibernate / power event:** `evidence/power_states.log` contains only 2 GSD keybinding-registration warnings; kernel journal shows no PM: suspend / Reached target Sleep
 - **Explicit unit-stop invocation** (`loginctl terminate-user`, `systemctl stop user@1000`, `KillUnit`, `Manager.StopUnit`): `evidence/stop_terminate_paths.log` contains only *post-kill* Removed session 1 and Removed session 4 from systemd-logind. No preceding stop RPC.
 - **CONST-033 host power-state transitions:** uptime intact (up 26 min at 15:34 → up 31 min at 15:38), no kernel suspend/hibernate messages
 - **Any process explicitly emitting kill -9 visible in journal:** no signal 9 to pid ... audit trail (audit rules absent — see below)
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The recurring architectural gap (unchanged across #2–#5)

auditd is not installed on this host: command `-v auditctl` returns nothing, command `-v ausearch` returns nothing, `/etc/audit/` does not exist, `/etc/audit/rules.d/` does not exist. The `docs/incidents/2026-08-19-sudo-audit-rules-for-operator.md` install script (Rev 3) was authored + committed but never run with `sudo`. Without those rules, the kernel audit subsystem cannot attribute a `signal=9` delivery to a specific initiator PID/UID/cmdline. `evidence/user_scope_events_head.log` and `evidence/kill_moment.log` prove the audit subsystem is running (type=1106/1113/1104 messages present), but the `SYSCALL kill` and `PATH exec loginctl|systemctl` watches that would name the initiator are absent from the rules file that does not exist.

Per `docs/incidents/2026-08-19-5th-forced-logout.md` §“Systematic-debugging Phase 4.5” (Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST + “if ≥ 3 fixes failed: STOP and question the architecture”), the architectural problem is not the SIGKILL mechanism — it is the audit-install gap that has now blocked Phase 1 attribution 5 consecutive times.

What is genuinely NEW in #5 vs #4

1. **Memory-consumption rate ×10 faster than the prior session** — 23.6 GB in 20:41 vs 25.1 GB in 3 h 22 min. The reboot did NOT reset the runaway consumer; whatever pattern of subagent-driven memory growth surfaced in prior sessions surfaced again immediately after a clean boot with the same wave-6 orchestration profile.
2. **Fastest recovery** — 13 s dead-time vs #4’s 13 min. This suggests operator was actively at the console and re-authenticated immediately.
3. **Container build activity in the pre-kill window** — `crun-buildah` scopes at 15:28:10-15 and `overlayfs xino` warnings across 15:27:14 → 15:28:15 (`evidence/kernel_events.log`). Not observed being called out for #2/#3/#4 timelines.
4. **Rate-limiter saturation** — the merge-search proxy issued dozens of 429 Too Many Requests responses in the seconds before the kill,

indicating heavy synthesised load from the scaling_probe pytest fleet.

None of these change the SIGKILL initiator attribution — all four are *correlate signals*, not causal proofs (§11.4.6).

Verdict (§11.4.6)

§11.4.6 Iron-Law status: STILL INCOMPLETE. Every observable class ruled out in #4 is ruled out again in #5. Every signature that fired in #4 fired in #5 in identical shape. The initiator remains unattributed because kernel audit rules are still not installed on the host. #5 matches #2/#3/#4 as one class of incident with high confidence: identical PAM close signature + identical bare SIGKILL exit shape + identical Linger=yes contradiction + identical absence of every candidate mechanism (kernel OOM, systemd-oomd, explicit stop RPC, power event). The load-and-memory correlation is stronger in #5 than in prior incidents but neither #4's nor #5's data can distinguish CAUSE from CORRELATION without the audit-rules attribution.

Evidence files (all under docs/qa/B0B-124/)

- journal_inc5.log — full journalctl 15:25-15:30 (11,330 lines)
- evidence/kill_moment.log — filtered kill window
- evidence/systemd_pid1_prekill.log — every systemd[1] message 15:28:15-22
- evidence/pre_kill_15s_window.log — 15-s pre-kill filtered journal
- evidence/signals_at_kill.log — GDM/mutter/logind/PAM signals at kill
- evidence/kernel_events.log — kernel messages (overlayfs, veth, rfskill; **no OOM**)
- evidence/kernel_oom_scan.log — **empty** (0 kernel-OOM hits)
- evidence/oomd_journal.log — systemd-oomd unit journal (Start only, no kill decisions)
- evidence/oom_extended.log — extended OOM/PSI/swap scan
- evidence/power_states.log — 2 hits (both false-positive keybinding registrations)
- evidence/stop_terminate_paths.log — logind session removals (all post-kill)
- evidence/gdm_compositor_events.log — GDM/gnome-shell activity around kill

- evidence/gdm_worker_3976.log — the actor of the PAM close (PID 3976 = gdm-session-worker)
- evidence/user1000_grepped.log — full user@1000.service lifecycle in window
- evidence/user1000_service_journal.log — helix-server journal noise (unrelated to kill)
- evidence/user_scope_events_head.log — user/session scope timeline (100-line head)
- evidence/loginctl_user_state.txt — Linger=yes State=active, Timestamp=15:07:40, IdleHint=no
- evidence/proc_pressure_memory.txt — current PSI (avg60=0.00 at investigation time)
- evidence/free_now.txt — memory state at investigation
- evidence/inc4_forensics_head.txt — first 80 lines of #4 forensics for comparison
- evidence/kernel_oom_scan.log — 0 hits, ruling out kernel OOM

Cross-references

- docs/incidents/2026-08-18-perceived-forced-logout-2nd.md (BOB-116, #2)
- docs/incidents/2026-08-18-3rd-forced-logout.md (BOB-120, #3)
- docs/incidents/2026-08-19-4th-forced-logout.md (BOB-123, #4)
- docs/incidents/2026-08-19-5th-forced-logout.md (BOB-124, #5)
- docs/incidents/2026-08-19-sudo-audit-rules-for-operator.md (Rev 3 — still uninstalled)
- docs/proposals/external-watchdog-for-forced-logout-architectural-gap.md (Path 2 proposal)
- ~/.claude-claude4/.../memory/forced_logout_incidents.md (playbook)

Iron Law compliance

No source changes made. No sudo invoked. No auditd installed. No shell scripts modified. Only evidence files written under docs/qa/BOB-124/.